Land (Soil) Pollution

The soil pollution is defined as the contamination caused by chemicals and other substances resulting in loss of the productivity or fertility of soil. The loss of productivity is in terms of yield of the crops. Besides the damaging contaminants, there are some other types of contaminants also do not cause any detrimental effect to plants, but create adverse effect on human beings and the animals who consume such plants.

Causes of Soil pollution

The main causes of soil pollution are:-

1. Poor management of solid and liquid waste disposal such as sewage, sludge, domestic waste, industrial waste and agricultural waste.
2. Use of chemical fertilizers and pesticides to raise the crop yields.
3. Improper sanitation methods.
4. Soil erosion
5. Salination and water logging.
6. Wetlands formed due to irrigational canals.
7. Acid rain
8. Disposal of medical waste.

Effects of soil pollution

The major effects of soil pollution as follows:-

1. Water mixed with human waste, sewage and sludge kills the micro – organism and reduces the fertility of soil.
2. The loss of fertility of soil leads to food shortage.
3. Garbage, sludge and other waste create bad odour which causes chronic diseases.
4. Pesticides seep gradually through the soil and contaminated the underground drinking water.
5. The urban solid waste obstructs the public passage.

Control of solid waste

The soil pollution can be controlled by following measures:-

1. By minimizing the generation of solid wastes.
2. By reusing and recycling of solid waste.
3. By employing proper solid waste disposal methods.
4. Methods should be adopted to reduce salination and water logging.
5. By tree plantation to reduce soil erosion.
6. By employing proper sanitation system.
7. By minimizing the use of chemical fertilizers, pesticides and other chemicals used for agricultural purposes.
8. By enforcing strict legislation and their effective implementation.

Solid waste management

The term solid waste management mainly refers to the complete process of collecting, treating and disposing of solid wastes. Solid wastes are those solids which are discarded and rendered useless by human beings. It also includes the animal wastes. The unwanted objects falls in the category of solid waste:-

1. The water material arise due to construction and demolition.

2. Ashes from industries, thermal power plants and other sources.
3. Rubbish including combustible and non – combustible solids.
4. Dead animals.
5. Garbage which includes organic waste and sludge

Causes of urban solid waste

The main causes for solid waste are:-

1. The advancement in technology leading to large scale production of consumable goods.
2. Growth in the consumption based society leading to enhancement of consumption items.
3. Use and throw culture which become possible due to availability of disposable items like containers, glass bottles cans etc.
4. Advancement of packaging technology due to which almost every item is packed before sale.
5. Increasing urbanization leading to generation of domestic solid waste and materials generated due to civil works.
6. Overpopulation is responsible for increased generation of garbage from each house.

Effects of solid waste

The main effects from solid waste are:-

1. Due to improper disposal of solid waste particularly by waste management organizations, the collected wastes gets heap up and become a problem for both the environment and also for the public.
2. By dumping of huge garbage, drives biodegradable materials to decay and decompose under abnormal, uncontrolled and unhygienic conditions.
3. The biodegradable waste becomes a breeding ground for different types of disease-causing insects as well as infectious organisms.
4. A foul smell is produced and it also spoils the aesthetic value of the area.
5. There is also aesthetic danger to the surroundings as the stray animals invade the dumping sites,
6. The crops and water supply may also get contaminated.
7. By burning the paper and other scraps along with the hazardous wastes, dioxins and poisonous gasses are produced and released into the air which results in causing various diseases including chronic disease, skin infections, cancer, etc.

Control Measures for solid wastes

Control of solid waste can be accomplished by following measures:-

1. By minimizing generation of domestic garbage and other wastes.
2. By throwing out the rubbish and other unwanted materials in proper place form where they may be easily taken for disposal.
3. By enforcing law in this regard.
4. By incorporating various methods of solid waste disposal.

Methods of solid waste disposal

The various methods for solid waste disposal are as follows:-

1. **Land filling**

In this method, solid wastes are dumped into low laying areas. The refuse is dumped in layers

of 1.5m height and each layer is covered by atleast 20 cm thick soil. Each layer is then left out for seven days and compaction is carried out for its settlement, before starting filling the next layer. Insecticides like DDT are sprayed on top to prevent breeding of mosquitos and flies. Generally the refuse get stabilized within 2 to 12 months and settle down by 20 % and 40% of its original height.

Advantages

1. Land filling is simple and economical.
2. Skilled labour is not required.
3. Low – lying areas can be reclaimed for better use.
4. No residue and by – products are formed and hence, no further is required,
5. No costly plants and equipment are required

Disadvantages

1. Continuous emission of foul smell near the site of disposal.
2. Use of insecticides is required.
3. Requirement of large land area.

2. Composting

In this method, the organic materials of solid waste are digested anaerobically and converted into stable material called humus. It is a nutrient-rich material that looks like dirt and can be used to improve soil and support plant growth. This humus can be used as organic fertilizers for agricultural purposes. It is a hygienic method carried out by anaerobic bacterial action. This method is suitable for Indian condition especially for small and medium sized town. It is the best method for recycling organic solid waste.

3. Incineration

Incineration means burning of solid waste in open hearth furnace. This method is generally used when suitable dumping land areas are not available. When solid waste is collected, the non – combustible and inert materials, such as, soil, broken glass, porcelain etc. separated so as to reduce the load on the furnace. The combustible garbage are burnt in furnace. The minimum temperature in combustion chamber is kept greater than 670^0C to incinerate all the organic waste and oxidizing foul smelling gases. The final products turn to ashes and clinkers, the ashes are used as aggregate for low grade concrete for road construction.

Advantages

1. There is no foul odour and dust problem arises.
2. All th pathogens and insects are destroyed.

Disadvantages

1. It has relatively high initial cost.
2. There may be problem of smoke, odour and ash during improper functioning of incinerators.

Air Pollution

Air Pollution is the release of pollutants such as gases, particles, biological molecules, etc. into the air that is harmful to human health and the environment.

Causes of air pollution

The major causes of air pollution are:-

1. Gases and dust particles release from burning of fossil fuel.
2. The gases and particles released from automobile exhaust.
3. Natural causes like volcanic eruption, dust storm and forest fire.
4. Smoke release from industries and domestic areas.
5. The gases and dust particles release from mining activities.
6. Chlorofluoro carbons (CFCs) release from air conditioner and refrigeration units.
7. Methane released into air due to decomposition of organic matter.
8. Burning of paddy straw.

Effects of air pollution

The major effects of air pollution are:-

1. The air pollution enhance the process of greenhouse effect due to release of gases, such as, CO_2 and methane, which leads to global warming. This global warming leads to climate change.
2. The release of oxides of carbon, nitrogen and sulphur causes acid rain.
3. The air pollution degrade the air quality index.
4. Air pollution may cause health problems, such as, respiratory problem, irritation of eyes, skin diseases, and irritation of throat.
5. The release of CFCs causes ozone layer depletion.
6. The air pollution also effect the animal health (aquatic as well as terrestrial animals).

Control of air pollution

The air pollution may be controlled by following measures:-

1. By minimizing the burning of fossil fuel.
2. By afforestation and tree plantation.
3. By using alternative for fossil fuels.
4. By encouraging the use of electric vehicle.
5. By using alternative for CFCs.
6. By using air pollution control equipment, such as, electrostatic precipitator, cyclone separator, spray tower and fabric filters.